

Spočítáme všechny mocniny matice  $A$  lineárního operátoru  $f$  a určíme jejich obory hodnot, tedy lineární obaly sloupců; a jádra, což jsou lineární obaly vektorů  $b_n$  zobrazujících se na 0, tedy těch, pro který má matice  $n$ -tý sloupec nulový:

Matice  $A$  lineárního operátoru  $f$  bude vypadat takto:

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

$$\text{Im}(f) = \text{Span}\{b_1, b_2, b_3, b_4, b_6, b_7, b_9\} \text{ a } \text{Ker}(f) = \text{Span}\{b_1, b_6, b_9\}.$$

$$A^2 = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\text{Im}(f^2) = \text{Span}\{b_1, b_2, b_3, b_6\} \text{ a } \text{Ker}(f^2) = \text{Span}\{b_1, b_2, b_6, b_7, b_9, b_{10}\}.$$

$$A^3 = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\text{Im}(f^3) = \text{Span}\{b_1, b_2\} \text{ a } \text{Ker}(f^3) = \text{Span}\{b_1, b_2, b_3, b_6, b_7, b_8, b_9, b_{10}\}.$$

$$A^4 = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$\text{Im}(f^4) = \text{Span}\{b_1\}$  a  $\text{Ker}(f^4) = \text{Span}\{b_1, b_2, b_3, b_4, b_6, b_7, b_8, b_9, b_{10}\}$ .

$\forall k \in \mathbb{N}, k \geq 5: A^k = 0_{10 \times 10}$ , proto  $\text{Im}(f^k) = \vec{0}$  a  $\text{Ker}(f^k) = \text{Span}(B)$ .

Díky těmto znalostem můžeme zapsat bázi prostoru  $\text{Ker}(f^i) \cap \text{Im}(f^j)$  pro všechna  $i, j \in \mathbb{N}$ :

| i \ j | 1                                   | 2                    | 3          | 4     | ≤5          |
|-------|-------------------------------------|----------------------|------------|-------|-------------|
| 1     | $b_1, b_6, b_9$                     | $b_1, b_6$           | $b_1$      | $b_1$ | $\emptyset$ |
| 2     | $b_1, b_2, b_6, b_7, b_9$           | $b_1, b_2, b_6$      | $b_1, b_2$ | $b_1$ | $\emptyset$ |
| 3     | $b_1, b_2, b_3, b_6, b_7, b_9$      | $b_1, b_2, b_3, b_6$ | $b_1, b_2$ | $b_1$ | $\emptyset$ |
| 4     | $b_1, b_2, b_3, b_4, b_6, b_7, b_9$ | $b_1, b_2, b_3, b_6$ | $b_1, b_2$ | $b_1$ | $\emptyset$ |
| ≤5    | $b_1, b_2, b_3, b_4, b_6, b_7, b_9$ | $b_1, b_2, b_3, b_6$ | $b_1, b_2$ | $b_1$ | $\emptyset$ |

A tedy i dimenzi  $\text{Ker}(f^i) \cap \text{Im}(f^j)$  pro všechna  $i, j \in \mathbb{N}$ :

| j \ i | 1 | 2 | 3 | 4 | 5+ |
|-------|---|---|---|---|----|
| 1     | 3 | 2 | 1 | 1 | 0  |
| 2     | 5 | 3 | 2 | 1 | 0  |
| 3     | 6 | 4 | 2 | 1 | 0  |
| 4     | 7 | 4 | 2 | 1 | 0  |
| 5+    | 7 | 4 | 2 | 1 | 0  |